



Thesis project proposal

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Projects Proposal: How Stable are Baselines: analysis on the Reliability of Reinforcement Learning Baselines

Potential title:

- How Stable are Baselines: analysis on the Reliability of Reinforcement Learning Baselines

Setting:

- Currently, there are a lot of baseline frameworks in RL to use. Conveniently for the user, these baselines provide stability, consistency, and reliability of (famous) algorithms within RL. One might assume that these baseline frameworks provide algorithms that perform equally well across different baselines, but is this the case? Each framework has its reasons for using, whether it is ease of use via just a few lines of code and a comprehensive API taking care of the rest, elaborately documented code for specific tasks and/or environments, or modularity along with ease of distributional learning options. However, these differences should not make a difference in performance when talking about specific algorithm implementations.

State-of-the-art:

- The current state-of-the-art shows a wide collection of baseline frameworks, covering many algorithms (also non-overlapping collections), and availability to the user from simple pipelines to modular building blocks which are easily adaptable and interchangeable. However, no work (as far as I know) has been done on comparing these baseline frameworks specifically. Each baseline framework does highlight other baseline frameworks on some of its attributes and/or flaws.

Remaining challenge: :

- The question that remains is whether the same algorithm across baseline frameworks performs consistently. Furthermore, we will look into how the relative ranking of algorithms is preserved across baseline frameworks. We will be looking into these questions, and compare them to the results of the original papers in the algorithms that have been published. Lastly, by answering these questions, we will fill a gap in the literature about direct comparisons of algorithms of baseline frameworks.

Solution:

- We will be doing an extensive literary search on baselines, algorithms, and environments to sort out what baselines are currently being used, what algorithms are provided within these baselines, what hyperparameter settings are used, what environments are used, how these settings align with the original papers, and whether we can directly compare results of the performance of algorithms. We will be experimenting with a fixed set of environments for a fair comparison of algorithms across different baseline frameworks to see whether the algorithm performance is similar across different baseline frameworks and whether relative ranking is preserved across different baselines.

Contribution:

- Our contribution is filling a gap in RL literature by providing knowledge about making comparisons across different baselines and different algorithms. This work should expose and highlight how certain comparisons of algorithms/baseline frameworks in some form may or may not be consistent, and how this conclusion is set. That is, given the implementation details and framework properties.

Experiment questions:

- To what extent do results reported in original papers align with those produced by existing baseline framework implementations?
- To what extent can results obtained from different baseline frameworks be considered directly comparable?
- To what extent is the relative performance ranking of algorithms preserved across different baseline frameworks?

Experiment design:

- We will be using a fixed set of environments for comparison. These environments are picked via popularity within the literature. Environments tested will not differ from repository. Environments may also be selected if major differences can be seen within reported performances across baselines, suggesting implementation details differ.
- We will start with one algorithm, PPO, that is widely applied. That way we can shape our hypothesis for remaining algorithms.
- We will use a number of baseline frameworks that are currently being used. This means these frameworks must be maintained, and include a number of (famous) algorithms. Good documentation and publication of an article is a must. This allows for further analysis within the framework to draw grounded conclusions.
- In terms of hyperparameters, we will be using hyperparameters of the original paper of the algorithm is applicable, else we will be applying the best performing hyperparameters of a single algorithm on the remaining algorithm for fair comparison. All experiments are repeated at least 5 times. Evaluations will be done in intervals of t timesteps. Each algorithm will run for a similar T timesteps total on a specific environment.

Timeline

Start date: February 19th
Projected first draft: July 15th
Projected final draft: August 15th
Projected thesis defense date: August 31st

Month	Plan	Product
Feb 19	Finish documents (formal proposal, contract), start with Thesis.	Project Proposal, contract
Mar 11	Looking into the different frameworks that are available. I will start with a vast selection of frameworks, and look into the related work as to what has motivated this framework to exist and what other frameworks exist. Making such an overview gives good insight and makes comparing frameworks more accessible.	I want to have a list of frameworks (that have implemented PPO) and a list of environments (where PPO has been experimented with via these frameworks) ready by this point.
Mar 25/Apr 1	Having the list of environments and baseline frameworks ready, I want to see whether I can run one iteration of an environment of each framework on an environment, to see how the frameworks' architecture works, how logging results work, and whether the list of frameworks must be adjusted. In the meantime, gathering an overview of the literature of these baseline frameworks and their results outlines its consistency among experiments even more.	I want to have an overview of the baseline frameworks' implementations, and possible difficulties as to how to implement each experiment accordingly. A time estimation as to how long each experiment may take would also be great to have by this point, such that further planning may be more efficiently done. Furthermore, the list of experimental results using baselines may grow continuously.
Apr 15	Set up the experiments on ALICE. I assume this will take some time and won't be as trivial, given the many versions and types of framework baselines. Given the ETA of the experiments, extending upon literary research is possible. However, up to this point we cannot decide yet whether to extend this to more algorithms, and thus additional literary research upon the reported performance on other algorithms is still 'redundant'.	Properly have everything set up on ALICE, and running. This includes having extensive loggers, using similar SLURM job attributes (such as CPUs/GPUs) for fair comparison. By this point, create a plan to extend this research to more overlapping algorithms from baselines,
Experiment 1	After the experiment with PPO applied on a fixed set of environments with implementations of PPO from several baselines, we can create a more thorough hypothesis and plan continuing onwards. This includes, but is not limited to, finding	A plan of the aforementioned proportions on how to continue with this research. This plan, similar to a plan as this one, would give more insight into the deadlines of

	out what causes these inconsistencies between implementations of PPO, extending our set of environments to be tested on, or finding potential other algorithms to look into RQ3	experiments to validate our research questions, the time spent on literary research of other algorithms, and whether we should change some attributes of our experiments.
Experiment 2+	I assume that everything worked out as planned, and that additional algorithms (as intended) can be looked into. Assuming that the setup on ALICE can be easily reused to be applied on other algorithms, given the implementations of the usage of each framework on PPO, makes setting up additional experiments not too difficult. However, by this time, there is a lack of literary research within the tested algorithms, which is then a must to fill this gap in the meantime.	Setting up additional experiments should be a fairly easy job. Running these whilst doing literary research within the algorithm makes for efficient execution. I cannot hold that the literary research is finished before the experiment.
15 Jun-Aug	Given the knowledge acquired during the time of experimenting, literary research and everything in between, I should be able to answer our three research questions. This means that in this period, I will be extensively writing my thesis. Before I start with this, I want to write up a plan as to the structure of the thesis, such that I do not get 'lost' or start being incohesive throughout the writing process. During this period, any unfinished experimenting (for whatever reason) may still be performed in the background, given the setup on ALICE remains properly accessible to adjustment to other algorithms. NEW algorithms should not be considered at this point, since literary research within this period interferes with the time set for writing the thesis.	Given a plan of a structure of the thesis, writing the thesis in parts should yield a draft by the end of July.
15 Aug -	Given the writing process, feedback of the first draft, and completeness of the thesis, the plan is to finetune the thesis based on this feedback and prepare for a thesis defense.	By this time, a complete thesis along with preparation for a thesis defense is met. Based on the thesis, we can discuss potential future steps